Bottleneck Migration in Hybrid Wired and Wireless Industrial Networks: A Discrete-Event Simulation Study for the *Matrix Factory Twin*

Breno da Silva Pereira Gabriel S. Stuart Lucas de Mello Kindermann

Department of Informatics and Statistics (INE)
Universidade Federal de Santa Catarina (UFSC)
Professor: Antônio Augusto Medeiros Fröhlich

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Abstract

Matrix-style manufacturing uses decoupled production islands linked by fleets of autonomous mobile robots (AMRs), coordinated through discrete-event digital twins such as the *Matrix Factory Twin*. Its communication infrastructure combines a wireless link between the AMR fleet and the fleet management system (FMS) with a wired Ethernet backbone shared by fixed manufacturing stations and telemetry services. This proposal presents an OMNeT++/INET simulation study of the traffic load and fleet-density conditions under which wired mechanisms (best-effort, IEEE 802.1Q priority, and IEEE 802.1Qbv TSN) and wireless mechanisms (contentious IEEE 802.11 access and an idealized scheduled regime) improve application timing. The study characterizes how the dominant timing bottleneck moves between the two network domains. Representative traffic classes are inspired by VDA 5050, DDS, OPC UA, and MQTT. The primary outcome is the rate of messages unavailable at their application consumption instant, supported by measurements of latency, jitter, loss, queue behavior, and network utilization. The results will form a reusable representation of network behavior for asynchronous consumption by the *Matrix Factory Twin*'s Next Event Request (NER) engine.

Keywords: Time-Sensitive Networking, IEEE 802.1Qbv, Scheduled Wireless Access, hybrid industrial networks, discrete-event simulation, OMNeT++/INET, digital twin, autonomous mobile robots.

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1 Introduction and Motivation

Matrix production organizes manufacturing around decoupled, reconfigurable process islands. Fleets of AMRs move material and work in progress between those islands, so production depends on coordination among vehicle scheduling, station availability, and network delivery. In the *Matrix Factory Twin*, a Next Event Request (NER) engine governs the simulation clock and advances it after all events with earlier timestamps have been resolved. Communication delays therefore affect both the fidelity of the twin and the time horizon over which its state can be evaluated.

The network that supports this coordination has two connected domains. IEEE 802.11 links the AMR fleet to the FMS, while an Ethernet backbone connects machining stations, access points, and telemetry services. IEEE 802.1Qbv time-aware shaping can bound and reduce contention-induced queuing delay and jitter in the wired domain through scheduled gate operation [2, 4]. The end-to-end result also depends on wireless access. Under high AMR density, IEEE 802.11 contention can introduce enough delay variation to obscure improvements achieved in the Ethernet backbone [5, 7]. With coordinated wireless access, Ethernet queuing can become dominant as background and telemetry traffic increase.

The study is based on the hypothesis that the effect of improving one domain depends on that domain's contribution to the critical end-to-end delays. Reducing wired variability should improve application timing when Ethernet queuing is significant. Reducing wireless contention should become more important as fleet density increases. As one contribution decreases, the other should account for a larger share of the observed delay. The transition is expected to vary gradually with traffic load and AMR fleet density.

The simulation campaign examines the operating conditions that produce this transition. It investigates when wireless contention neutralizes the timing benefit of IEEE 802.1Qbv, when Ethernet switching delay becomes dominant under coordinated wireless access, and which combinations of background load and fleet density require deterministic behavior in both domains to satisfy the application timing requirements. These questions define the comparisons and measurements presented in Section 3.

The scope is limited to queuing dynamics, medium contention, and traffic coexistence under load. The topology remains fixed and operational during every run, excluding access point outages, cable breaks, handoffs, and other structural changes. The wireless environment is controlled so that fleet density captures the effect of contention and coordinated access. Detailed propagation and mobility models are excluded. The scheduled wireless regime is a conflict-free reference for coordination; detailed IEEE 802.11ax OFDMA signaling and contention are outside the present scope [6]. Network results are produced independently of the evolving physical state of the factory and are later represented for use by the twin. These assumptions keep the experiment feasible within the six-week schedule in Section 4.

2 Objectives

2.1 General Objective

To experimentally investigate, in a discrete-event simulation environment, the operational boundaries of load and fleet density at which local improvements in the Ethernet and wireless domains translate into deadline conformance for the *Matrix Factory Twin*, and to map how the timing bottleneck migrates between the two domains as those boundaries are crossed.

2.2 Specific Objectives

1. Model synthetic traffic profiles inspired by VDA 5050 (AMR orders and state reports), DDS (inter-station coordination), and OPC UA/MQTT (background telemetry) [8, 9, 10, 11].
2. Build the hybrid simulation infrastructure in OMNeT++/INET. Three wired mechanisms

will be implemented: best-effort, IEEE 802.1Q class-based priority, and IEEE 802.1Qbv TSN with a Time-Aware Shaper. Two wireless mechanisms will be compared: contentious IEEE 802.11 CSMA/CA and the idealized scheduled regime described in Section 1 [1, 2, 5].

3. Compare the six backbone and radio combinations under variations in background Ethernet load and AMR fleet density, using application deadline adherence and complementary network-performance indicators.
4. Identify, from the resulting response surfaces, the load and density thresholds at which the dominant bottleneck migrates from one domain to the other and determine when coordinated improvements are required in both domains.
5. Generate multidimensional lookup tables containing the empirical delay distributions and loss probabilities produced by the experimental campaign, structured for asynchronous consumption by the *Matrix Factory Twin*'s NER engine.

3 Methodology and Experimental Design

3.1 Simulation Environment and Tooling

The study will be carried out in OMNeT++ with the INET framework, using the TSN capabilities provided natively by current INET releases, including IEEE 802.1Qbv time-aware shaping and gate scheduling [3, 4]. The simulated topology consists of Ethernet switches interconnecting machining stations, one or more wireless access points, and background telemetry sources, all converging on a fleet-management server (FMS); the AMR fleet communicates with the FMS exclusively over the wireless link.

3.2 Comparative Scenarios: Backbone and Radio Matrix

The core of the study is the comparison of three wired-backbone mechanisms with two wireless-access regimes, producing the six scenarios summarized in Table 1. This structure makes it possible to observe whether a local improvement remains visible at the application level when the other network domain is unchanged.

Table 1: Backbone and radio comparison matrix.

Wired Backbone	**IEEE 802.11 CSMA/CA**	**Idealized Scheduled Access**
Best-effort (single FIFO)	Cell 1	Cell 2
IEEE 802.1Q priority queuing	Cell 3	Cell 4
IEEE 802.1Qbv TSN (TAS/GCL)	Cell 5	Cell 6

The wired scenarios represent progressively stronger forms of traffic differentiation: ordinary best-effort Ethernet, IEEE 802.1Q class-based priority, and IEEE 802.1Qbv time-aware scheduling [1, 2]. The wireless scenarios include contention-based IEEE 802.11 access and an idealized coordinated regime. The coordinated regime represents conflict-free scheduled access within the scope defined in Section 1; detailed IEEE 802.11ax procedures are outside the model [5, 6].

3.3 Wireless-to-Wired Synchronization and Gate-Admission Delay

The transition between the wireless and wired domains is part of the phenomenon under study. Traffic arriving from the radio may not be aligned with the transmission opportunities offered by IEEE 802.1Qbv, so TSN cannot be assumed to eliminate every source of delay [2, 4]. The model

will therefore preserve the effect of admission into the scheduled wired domain and distinguish it from delays accumulated over the wireless link. Time synchronization is treated as an enabling assumption for the TSN scenario; a detailed validation of synchronization across the wireless segment is outside the scope of this project.

3.4 Traffic Characterization and QoS Classes

The simulation uses representative traffic classes inspired by the communication roles and messaging models of VDA 5050, DDS, OPC UA, and MQTT [8, 9, 10, 11]. Their timing criticality, traffic intensity, and application consumption instants are defined as experimental modeling parameters. The standards provide the communication context for each class, and synthetic generators reproduce the traffic needed for the evaluation.

Table 2: Traffic classes and their role in the study.

Traffic Class	**Reference**	**Role in the Evaluation**
AMR commands and dispatch orders	VDA 5050	Timing-sensitive control traffic sent toward mobile resources.
Vehicle state reports	VDA 5050	Recurrent wireless traffic representing the operational state of the AMR fleet.
Inter-station coordination	DDS	Periodic industrial coordination traffic with sensitivity to delay variation.
Background telemetry	OPC UA and MQTT	Non-critical traffic that competes for wired-network resources and creates coexistence conditions.

3.5 Treatment of Delay and Loss

A message may become unavailable to the application because it is lost or because it arrives after its intended consumption instant. The evaluation will preserve this distinction and will also examine whether the delay originates primarily in wireless contention, wired queuing or scheduling, or recovery mechanisms. Exact protocol parameters will be defined during implementation and documented in the final experimental report.

3.6 Primary Metric: Application Deadline-Miss Rate

The primary outcome is the proportion of messages that are unavailable when the application is expected to consume them. A message is considered a miss when it is lost or arrives after the relevant digital-twin tick. This application-oriented metric allows the study to compare network configurations according to their practical effect on the *Matrix Factory Twin*, while retaining the underlying cause of each miss for interpretation.

3.7 Secondary Metrics and Instrumentation

The primary metric will be interpreted alongside end-to-end latency, delay variation, packet loss, queue behavior, and utilization in the wired and wireless segments. These indicators will attribute deadline misses to the relevant domain and show the end-to-end effect of each network configuration.

3.8 Experimental Campaign and Statistical Validity

For each scenario in Table 1, the evaluation will vary background Ethernet load and AMR fleet density over representative operating ranges. Independent replications and uncertainty analysis will be used to distinguish systematic behavior from simulation variability [12]. The exact levels, number of replications, and run duration will be established after a pilot phase, balancing statistical confidence with the six-week execution window. The resulting comparisons

will support the identification of operating regions in which the dominant timing bottleneck changes domain.

3.9 Output Structure: Lookup Tables and Interface with the NER Engine

The simulation results will be consolidated into a structured representation of network behavior organized by backbone configuration, wireless regime, traffic conditions, fleet density, and traffic class. Each operating region will summarize the observed delay and loss behavior. The *Matrix Factory Twin*'s NER engine can then reproduce the measured network effects through asynchronous access to this representation.

4 Timeline

This project follows a compressed, four-phase course schedule (Table 3). The six-week window defines a scope based on an idealized scheduled wireless reference, open-loop integration with the factory twin, and a controlled wireless environment. The experimental campaign focuses on the bottleneck migration produced by traffic load and AMR fleet density.

Table 3: Execution timeline (P1 to P4), activities, and deliverables.

Phase	**Due Date**	**Main Activity**	**Deliverable**
P1	Oct. 26, 2026	Literature review; finalization of the hypothesis, scope, modeling assumptions, base topology, and representative traffic classes.	Approved proposal and validated base simulation scenario.
P2	Nov. 9, 2026	Implementation and integration of the wired and wireless comparison scenarios.	Operational hybrid scenario covering the six combinations in Table 1.
P3	Nov. 23, 2026	Execution of the comparative campaign, analysis of application deadline adherence, and preliminary mapping of bottleneck migration.	Consolidated results and preliminary interpretation.
P4	Dec. 7, 2026	Organization of the network-behavior representation, interpretation of the operating regions, and preparation of the final communication materials.	Final report, supporting data representation, and presentation.

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